



EpiAwareNet

Epigenome-aware Transformer for Gene Regulatory Network Inference
from Single-Cell Multi-Omics Data

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EpiAwareNet_pub
KDD

Motivation

Gene regulatory networks (GRNs) explain how transcription factors (TFs) control the genes that define a cell's identity. Single-cell data now profile both **gene expression** (scRNA-seq) and **chromatin accessibility** (scATAC-seq) in the same cells, yet most GRN methods use expression alone and ignore the **epigenomic context** that determines whether a TF can act.

Challenges we address:

- ▶ Regulatory logic is **cell-context specific** — one static network cannot fit all cell states.
- ▶ Ground-truth edges are scarce and one-sided: we observe **positives only**, never confirmed negatives.
- ▶ Coupling expression with distal accessible peaks requires **scalable** attention over thousands of genes and peaks.

Key Contributions

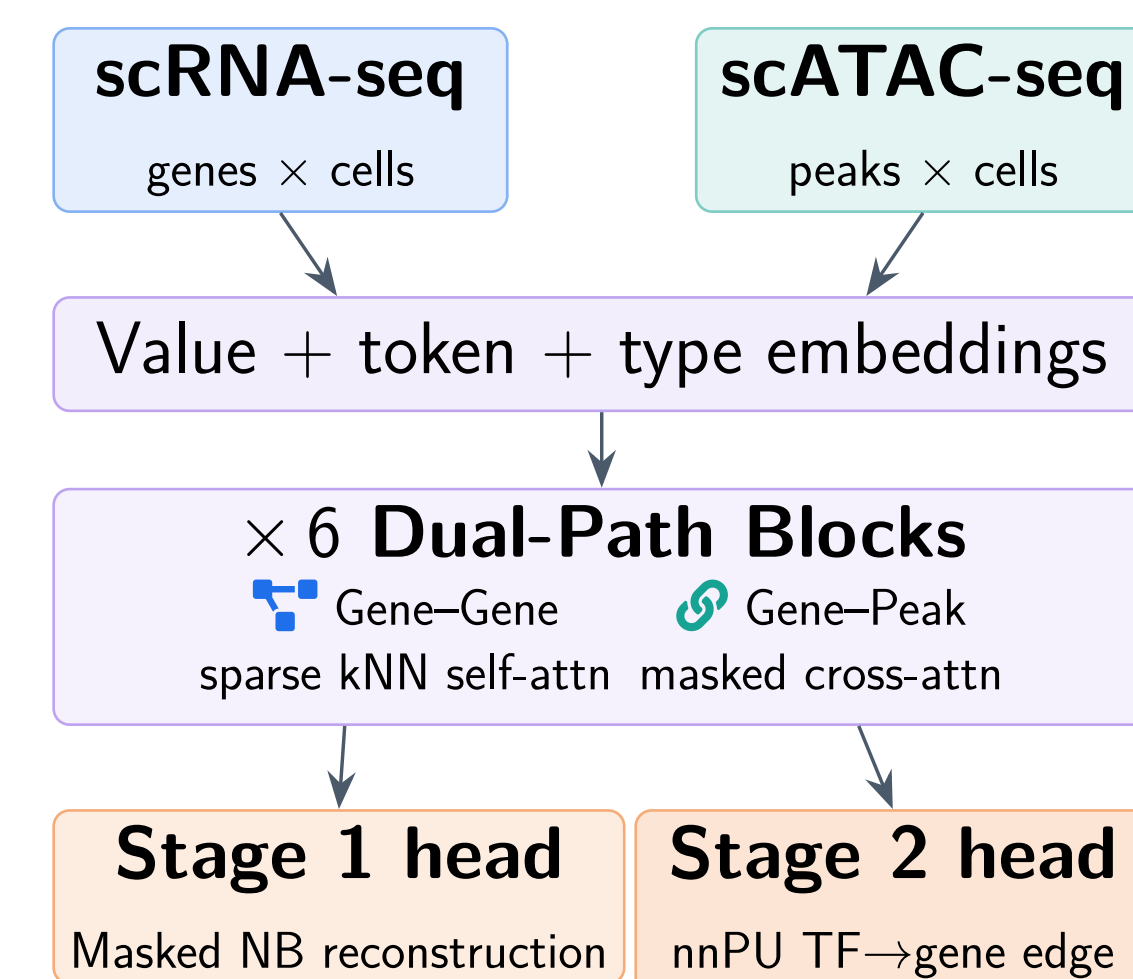
- ▶ A **dual-path Transformer backbone** that fuses RNA and ATAC through gene–gene sparse self-attention and candidate-aware gene–peak cross-attention.
- ▶ A self-supervised **masked negative-binomial** objective that respects the count nature of single-cell expression.
- ▶ A **non-negative PU** (nnPU) fine-tuning head that learns TF→gene edges from positive priors *without* fabricated negatives.
- ▶ **Context-specific GRNs**: per-cell scoring aggregated over any cell subset yields a network for that state.

At a Glance

2 omics layers
2 training stages
 $O(nk)$ sparse attention

Inputs: scRNA & scATAC matrices, gene→candidate-peak map, known TF list, bulk positive TF–target links.

Model Architecture



The backbone alternates **gene–gene sparse self-attention** (each gene attends only to its k nearest neighbours in embedding space, cost $O(nk)$ not $O(n^2)$) and **gene–peak cross-attention** restricted to each gene's candidate peaks with optional Top- K routing. Stage 1 pre-trains the shared backbone; Stage 2 freezes it and trains only a light edge head.

Two-Stage Learning

Stage 1 — Self-supervised representation. Randomly mask a fraction of gene counts and reconstruct them with a **negative-binomial** likelihood, capturing the over-dispersed, integer nature of scRNA-seq:

$$\mathcal{L}_{\text{NB}} = -\sum_{i \in \mathcal{M}} \log \text{NB}(x_i \mid \mu_i, \theta_i),$$

with per-gene mean μ_i and dispersion θ_i predicted from the fused embedding (masked positions only).

Stage 2 — nnPU edge learning. The backbone is frozen; a head scores a [TF || gene] pair. With positive prior π_p and hinge loss ℓ , the **non-negative PU risk** is

$$R = \pi_p R_p^+ + \max(0, R_u^- - \pi_p R_p^-),$$

so unlabeled pairs act as noisy negatives without ever asserting a false negative — well suited to the incomplete gold standard.

Design Notes & Scalability

- ▶ **Sparse over dense.** Gene self-attention is restricted to k nearest neighbours (cosine kNN, FAISS-optional), turning the quadratic $O(n^2)$ cost into $O(nk)$ over thousands of genes.
- ▶ **Biology-guided sparsity.** Cross-attention only sees a gene's candidate peaks, injecting the prior that regulation acts through nearby accessible DNA.
- ▶ **Chunked attention** over genes/peaks keeps memory flat, so the same model scales across cell, gene, and peak counts.
- ▶ **Frozen backbone reuse.** One pre-trained encoder serves the reconstruction and edge tasks — Stage 2 adds only a light head.

Context-Specific GRNs

After fine-tuning, EpiAwareNet scores every TF–gene pair **per cell**. Averaging edge probabilities across any chosen cell subset $\mathcal{C}$ yields a context adjacency

$$A_{t \rightarrow g}^{\mathcal{C}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} d(f(z_t^c, z_g^c)),$$

so a single trained model produces different networks for different cell types, states, or conditions.

Evaluation Protocol

Held-out bulk TF–target links serve as the gold standard. We report **AUPRC** and **AUROC** with PR/ROC curves, plus a **degree-matched random-network** permutation test that yields per-TF precision/recall and **empirical p -values** — controlling for hub genes that inflate naïve overlap.

Datasets	pN, mN, PBMC, mouse brain
Metrics	AUPRC, AUROC, per-TF p -value
Gold	bulk TF–target priors

Ablation Studies

The released suite isolates each design choice:

- ▶ **Multi-omic vs. RNA-only** backbone — quantifies the value of chromatin accessibility.
- ▶ **nnPU vs. BCE** — PU learning vs. treating unlabeled pairs as hard negatives.
- ▶ **Cross-attention Top- K** $\in \{0, 2, 4, 8, 16\}$ — how much peak routing to allow.
- ▶ **Prior-noise & weak-prior** robustness — random candidate peaks and corrupted positive links.

Takeaways

- ✓ Epigenome-aware attention couples expression with accessible regulatory DNA.
- ✓ PU learning matches the positive-only reality of GRN supervision.
- ✓ One model, many **context-specific** networks.

Code, pipelines & ablations: github.com/tianyang-x/EpiAwareNet_pub | MIT License

End-to-End Pipeline

